

AZIMBEK KHUDOYBERDIEV

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 Azimbek  SUNY Korea  AI2S Lab  IDCITI

Career Objective

Career Objective  I develop adaptive deep learning models for multimodal sensor fusion to enable robust positioning and localization in autonomous systems, bridging vision, time-frequency signal processing, and efficient computing for reliable real-time state estimation in GNSS-challenged environments.

Education

2020.09 – Present  **Ph.D. Candidate, Computer Science** Stony Brook University (SUNY Korea Campus)
2018.09 – 2020.08  **M.Sc. Department of Computer Engineering**, Jeju National University, Korea
2014.09 – 2018.07  **B.Sc. Information Technology**, Tashkent University of Information Technologies, Uzbekistan

Work

2023.03 – Present  **AI Engineer**, IDCITI, Incheon, Korea

- Implementation and Deployment of AI-based solutions for smart transportation and positioning systems.

Research

Ph.D. Researcher  AI2S Lab, Department of CS, SUNY Korea.
Master Researcher  Jeju National University, Department of Computer Engineering, IoT and Mobile Computing Research Laboratory

Teaching

2021.08 – 2022.08  **Lecturer**. Kimyo International University in Tashkent. Department of Applied Informatics.

- Fall 21: Computer Networks, Internet of Things and Embedded Systems
- Spring 22: AI, Data Science

Spring 21, 24, 25  **Graduate TA**. Department of Computer Science, SUNY Korea

- CSE 101: Computer Science Principles (Spring 21, Spring 24),
- CSE 310: Computer Networks (Spring 25),
- CSE 220: System Fundamentals I (Spring 25).

Skills

Programming & Scripting	Python, C/C++, MATLAB, Java, Bash, ROS
Deep Learning & AI Frameworks	PyTorch, TensorFlow, OpenCV, YOLO, Scikit-learn, and so on.
Deep Learning Architectures	Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Graph Neural Networks (GNN), Long Short-Term Memory (LSTM), Transformer, Attention Mechanisms, Autoencoders
IoT & Embedded Systems	Experience in designing and developing various solutions involving multi-modal sensor integration (GNSS, IMU, Odometry, LiDAR, Cameras) and wireless communication protocols (BLE, Wi-Fi, MQTT) for smart environments and positioning applications.

Conference Acceptances

- Accepted
- IoN GNSS+ 2026:** Pseudolite at its Finest: Comprehensive Performance Report of uGPS, GNSS Signal Generation in an Active Tunnel. A. Khudoyberdiev, H. Y. Kim, and J. Ryoo.
 - IoN GNSS+ 2026** GFlowLoc: GFlowNet-Driven Adaptive Fusion for Robust Positioning in GNSS-Challenged Environments. A. Khudoyberdiev, and J. Ryoo.

Research Publications

Journal Articles

- 1 K. P. Ismaylova, A. Khudoyberdiev, and H.-C. Kim, "Cognimoe: End-to-end multimodal mental workload classification via on-the-fly scalogram generation and moe gating," *IEEE Sensors Journal*, pp. 1–1, 2025. [DOI: 10.1109/JSEN.2025.3640827](#).
- 2 A. Khudoyberdiev, H. Young Kim, and J. Ryoo, "Plus-code+: Zero-installment rover indoor localization," *IEEE Sensors Journal*, vol. 25, no. 12, pp. 23 088–23 104, 2025. [DOI: 10.1109/JSEN.2025.3561331](#).
- 3 H. Kim, A. Khudoyberdiev, S. S. R. Garnaik, A. Bhattacharya, and J. Ryoo, "Cloud-codec: A new way of storing traffic camera footage at scale," *ACM Trans. Multimedia Comput. Commun. Appl.*, vol. 21, no. 8, Aug. 2025, ISSN: 1551-6857. [DOI: 10.1145/3744649](#).
- 4 A. Khudoyberdiev, S. S. Garnaik, H. Y. Kim, and J. Ryoo, "A study of the man and unmanned teaming system for improving efficiency in a fulfillment center," *IEEE Access*, vol. 11, pp. 139 587–139 600, 2023.
- 5 A. Khudoyberdiev, M. A. Jaleel, I. Ullah, *et al.*, "Enhanced water quality control based on predictive optimization for smart fish farming," *Computers, Materials & Continua*, vol. 75, no. 3, 2023.
- 6 A. Khudoyberdiev, I. Ullah, and D. Kim, "Optimization-assisted water supplement mechanism with energy efficiency in iot based greenhouse," *Journal of Intelligent & Fuzzy Systems*, vol. 40, no. 5, pp. 10 163–10 182, 2021.
- 7 S. Ahmad, F. Jamil, A. Khudoyberdiev, and D. Kim, "Accident risk prediction and avoidance in intelligent semi-autonomous vehicles based on road safety data and driver biological behaviours," *Journal of Intelligent & Fuzzy Systems*, vol. 38, no. 4, pp. 4591–4601, 2020.
- 8 W. Jin, A. Khudoyberdiev, and D. Kim, "Risk assessment inference approach based on geographical danger points using student survey data for safe routes to school," *IEEE Access*, vol. 8, pp. 180 955–180 966, 2020.
- 9 A. Khudoyberdiev, S. Ahmad, I. Ullah, and D. Kim, "An optimization scheme based on fuzzy logic control for efficient energy consumption in hydroponics environment," *Energies*, vol. 13, no. 2, p. 289, 2020.

- 10 S. Ahmad, A. Khudoyberdiev, and D. Kim, "Towards the task-level optimal orchestration mechanism in multi-device multi-task architecture for mission-critical iot applications," *IEEE Access*, vol. 7, pp. 140 922–140 935, 2019.
- 11 A. Khudoyberdiev, W. Jin, and D. Kim, "A novel approach towards resource auto-registration and discovery of embedded systems based on dns," *Electronics*, vol. 8, no. 4, p. 442, 2019.

Complete Publication list on Google Scholar